

Annex 93A

(normative)

Specification methods for electrical channels

93A.1 Channel Operating Margin

Change Table 93A–1 as follows (some unchanged rows not shown):

Table 93A–1—COM parameters

Parameter	Reference	Symbol	Units
Signaling rate	93A.1.1	f_b	GBd
Maximum start frequency	93A.1.1	$f_{\min}$	GHz
Maximum frequency step	93A.1.1	Δf	GHz
Device package model Single-ended device capacitance ^a <u>Single-ended device series inductance^a</u> <u>Single-ended bump capacitance^a</u> Transmission line length ^a Single-ended package capacitance at package-to-board interface ^a Transmission line characteristic impedance ^a <u>Transmission line 2 length^a</u> <u>Transmission line parameter, a_1^b</u> <u>Transmission line parameter, a_2^b</u> <u>Transmission line 2 characteristic impedance^a</u>	93A.1.2	C_d L_s C_b z_p C_p Z_c z_{p2} a_1 a_2 Z_{c2}	nF nH nF mm nF Ω mm ns ^{1/2} /mm ns/mm Ω
Single-ended reference resistance	93A.1.2	R_0	Ω
Single-ended termination resistance	93A.1.3	R_d	Ω
Receiver 3 dB bandwidth	93A.1.4.1	f_r	GHz
Transmitter equalizer, minimum cursor coefficient	93A.1.4.2	$c(0)$	—
<u>Transmitter equalizer, 3rd pre-cursor coefficient^c</u> <u>Minimum value</u> <u>Maximum value</u> <u>Step size</u>	93A.1.4.2	<u>$c(-3)$</u>	 = = =
Transmitter equalizer, 2 nd pre-cursor coefficient ^c Minimum value Maximum value Step size	93A.1.4.2	$c(-2)$	 — — —
Transmitter equalizer, pre-cursor coefficient Minimum value Maximum value Step size	93A.1.4.2	$c(-1)$	 — — —

Table 93A-1—COM parameters (*continued*)

Parameter	Reference	Symbol	Units
Transmitter equalizer, post-cursor coefficient	93A.1.4.2	$c(1)$	—
Minimum value			—
Maximum value			—
Step size			—
...			
Decision feedback equalizer (DFE) length	93A.1.6	N_b	UI
Normalized DFE coefficient magnitude limit ^d	93A.1.6	$b_{\max}(n)$	—
Normalized DFE coefficient maximum limit ^d	93A.1.6	$bb_{\max}(n)$	==
Normalized DFE coefficient minimum limit ^d	93A.1.6	$bb_{\min}(n)$	==
Number of DFE floating tap banks ^d	93A.1.6	N_{bg}	==
Number of DFE floating taps per bank ^d	93A.1.6	N_{bf}	==
DFE maximum span including floating taps ^d	93A.1.6	N_f	UI
Normalized coefficient magnitude limit for DFE floating taps ^d	93A.1.6	$b_{g\max}$	==
DFE floating tap tail root-sum-of-squares limit ^d	93A.1.6.1	σ_{imax}	==
DFE floating tap tail starting position ^d	93A.1.6.1	N_{ts}	==
Random jitter, RMS	93A.1.6	σ_{RJ}	UI
...			

^aSome clauses that invoke this method do not provide a value for C_d , L_s , C_{b1} , z_{p1} , C_{p2} , Z_{c2} , z_{p2} , or Z_{c2} . See 93A.1.2.

^bSome clauses that invoke this method do not provide a value for a_1 or a_2 . See 93A.1.2.3.

^cSome clauses that invoke this method do not provide a value for $c(-2)$ or $c(-3)$. See 93A.1.6.

^dSome clauses that invoke this method do not provide a value for $b_{\max}(n)$, $bb_{\max}(n)$, $bb_{\min}(n)$, N_{bg} , N_{bf} , N_f , $b_{g\max}$, σ_{imax} , or N_{ts} . See 93A.1.6.

Change Table 93A–2 as follows (some unmodified rows are not shown):

Table 93A–2—Physical Layer specifications that employ COM

Physical Layer	Parameter values
...	
100GBASE-CR2 (Clause 136)	Table 136–18
100GBASE-CR1 (Clause 162)	Table 162–20
100GBASE-KR4 (Clause 93)	Table 93–8
100GBASE-KP4 (Clause 94)	Table 94–17
100GBASE-KR2 (Clause 137)	Table 137–6
100GBASE-KR1 (Clause 163)	Table 163–11
CAUI-4 C2C (Annex 83D)	Table 83D–6
100GAUI-4 C2C (Annex 135D)	Table 83D–6
100GAUI-2 C2C (Annex 135F)	Table 120D–8
100GAUI-1 C2C (Annex 120F)	Table 120F–8
200GBASE-CR4 (Clause 136)	Table 136–18
200GBASE-CR2 (Clause 162)	Table 162–20
200GBASE-KR4 (Clause 137)	Table 137–6
200GBASE-KR2 (Clause 163)	Table 163–11
200GAUI-8 C2C (Annex 120B)	Table 83D–6 ^a
200GAUI-4 (Annex 120D)	Table 120D–8
200GAUI-2 C2C (Annex 120F)	Table 120F–8
400GBASE-CR4 (Clause 162)	Table 162–20
400GBASE-KR4 (Clause 163)	Table 163–11
400GAUI-16 (Annex 120B)	Table 83D–6 ^a
400GAUI-8 (Annex 120D)	Table 120D–8
400GAUI-4 C2C (Annex 120F)	Table 120F–8

^aWith the exceptions given in 120B.4.

93A.1.2 Transmitter and receiver device package models

93A.1.2.1 Cascade connection of two-port networks

Change the last sentence in 93A.1.2.1 as follows:

For the purpose of this annex, this set of operations is referred to using the shorthand notation $S^{(z)} = \text{cascade}(S^{(x)}, S^{(y)})$. The shorthand notation $S^{(z)} = \text{cascade}(S^{(w)}, S^{(x)}, S^{(y)})$ is equivalent to $S^{(z)} = \text{cascade}(S^{(w)}, \text{cascade}(S^{(x)}, S^{(y)}))$. Following this approach, the notation can be extended to an arbitrary number of components.

93A.1.2.2 Two-port network for a shunt capacitance

Change the second paragraph in 93A.1.2.2 as follows:

The scattering parameters for the device capacitance C_d are denoted as $S^{(d)}=S(C_d)$. the scattering parameters for the device-to-package interface capacitance are denoted as $S^{(b)}=S(C_b)$, and the scattering parameters for the board capacitance C_p are denoted as $S^{(p)}=S(C_p)$. When a value for C_b is not provided by the clause that invokes this method, C_b is set to 0.

Insert 93A.1.2.2a after 93A.1.2.2:

93A.1.2.2a Two-port network for a series inductance

The scattering parameters for a series inductance with value L are defined by Equation (93A–9a).

$$S(L) = \frac{1}{2 + (j\omega L)/R_0} \begin{bmatrix} (j\omega L)/R_0 & 2 \\ 2 & (j\omega L)/R_0 \end{bmatrix} \quad (93A-9a)$$

The scattering parameters for the series inductance L_s are denoted as $S^{(s)}(L_s)$. When a value for L_s is not provided by the clause that invokes this method, L_s is set to 0.

93A.1.2.3 Two-port network for the package transmission line

Change the first paragraph of 93A.1.2.3 as follows:

The scattering parameters for the package transmission line model are a function of the complex propagation coefficient defined by Equation (93A–9), Equation (93A–10), and Equation (93A–11) and the reflection coefficient defined by Equation (93A–12). The values of the parameters that appear in these equations are defined in Table 93A–3 unless alternative values are provided by the clause that invokes this method. Where a value for Z_c is not provided by the clause that invokes this method, it takes the value 78.2 Ω . The units of f are GHz.

Insert the following at the end of 93A.1.2.3:

For clauses that use a second package transmission line segment described by parameters z_{p2} and Z_{c2} , the scattering parameters for the second transmission line are defined by Equation (93A–12), Equation (93A–13), and Equation (93A–14) where values for z_p and Z_c in the equations are provided by z_{p2} and Z_{c2} .

Change the title of Table 93A–3 as follows:

Table 93A–3—Transmission line model parameters and default values

93A.1.2.4 Assembly of transmitter and receiver device package models

Change 93A.1.2.4 as follows:

The scattering parameters for the transmitter device package model $S^{(tp)}$ are the result of the cascade connection of the ~~device capacitance model~~, package transmission line, and board capacitance as shown in Figure 93A–2 and defined by Equation (93A–15) and Equation (93A–15a).

$$\cancel{S^{(tp)} = \text{cascade}(\text{cascade}(S^{(d)}, S^{(l)}), S^{(p)})} \quad \cancel{S^{(td)} = \text{cascade}(S^{(d)}, S^{(s)}, S^{(b)})} \quad (93A-15)$$

$$\underline{S^{(tp)} = \text{cascade}(S^{(td)}, S^{(l)}, S^{(p)})} \quad (93A-15a)$$

Similarly, the scattering parameters for the receiver device package model $S^{(rp)}$ are the result of the cascade connection of the board capacitance, package transmission line, and ~~device capacitance model~~ as shown in Figure 93A–2 and defined by Equation (93A–16) and Equation (93A–16a).

$$\cancel{S^{(rp)} = \text{cascade}(\text{cascade}(S^{(p)}, S^{(l)}), S^{(d)})} \quad \cancel{S^{(rd)} = \text{cascade}(S^{(b)}, S^{(s)}, S^{(d)})} \quad (93A-16)$$

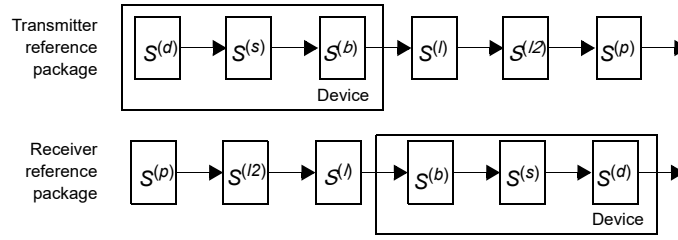
$$\underline{S^{(rp)} = \text{cascade}(S^{(p)}, S^{(l)}, S^{(rd)})} \quad (93A-16a)$$

The transmitter and receiver device package model scattering parameters for clauses that include a second transmission line segment are computed using Equation (93A–16b) and Equation (93A–16c), respectively.

$$\underline{S^{(tp)} = \text{cascade}(S^{(td)}, S^{(l)}, S^{(l2)}, S^{(p)})} \quad (93A-16b)$$

$$\underline{S^{(rp)} = \text{cascade}(S^{(p)}, S^{(l2)}, S^{(l)}, S^{(rd)})} \quad (93A-16c)$$

Insert Figure 93A–2 at the end of 93A.1.2.4 as follows:



NOTE— For some PMDs, $S^{(l/2)}$ is not included in the package model.

$S^{(d)}$ = device capacitance S-parameter
 $S^{(s)}$ = device series inductance S-parameter
 $S^{(b)}$ = bump capacitance S-parameter
 $S^{(l)}$ = package transmission line S-parameter
 $S^{(l/2)}$ = package transmission line 2 S-parameter
 $S^{(p)}$ = package capacitance S-parameter

Figure 93A–2—Reference package models

93A.1.6 Determination of variable equalizer parameters

Change the first paragraph of 93A.1.6 as follows:

COM is a function of the variables $c(-3)$, $c(-2)$, $c(-1)$, $c(1)$, g_{DC} , and g_{DC2} . The following procedure is used to determine the values of these variables that are used to calculate COM.

- Compute the pulse response $h^{(k)}(t)$ of each signal path k for a given $c(-3)$, $c(-2)$, $c(-1)$, $c(1)$, g_{DC} , and g_{DC2} using the procedure defined in 93A.1.5.
- Define t_s to be the time that satisfies Equation (93A–25). If there are multiple values of t_s that satisfy the equation, then the first value prior to the peak of $h^{(0)}(t)$ is selected. The ~~fixed-tap~~ coefficients of the ~~decision feedback equalizer~~ DFE $b(n)$ are computed as shown in Equation (93A–26). If N_b is 0, then the $b(n)$ is considered to be zero for all n . If $bb_{\min}(n)$ and $bb_{\max}(n)$ are not provided by the clause that invokes this method then $bb_{\min}(n)$ is set to $-b_{\max}(n)$ and $bb_{\max}(n)$ is set to $b_{\max}(n)$.
- Compute the location and values of floating-tap coefficients of the DFE to minimize the residual ISI using the method described in 93A.1.6.1. If N_{bg} , N_{bf} , N_f , and $b_{g\max}$ are not specified then no floating taps are used and N_f takes the value of N_b from the referring clause.
- ~~d)~~ Define A_s to be $R_{LM}h^{(0)}(t_s)/(L-1)$.
- ~~e)~~ Compute σ_{TX}^2 per Equation (93A–30) and Equation (93A–29). This represents the noise output from the transmitter.
- ~~f)~~ Compute $h_{ISI}(n)$ per Equation (93A–27). This represents the residual intersymbol interference (ISI) after decision feedback equalization. The corresponding ISI amplitude variance σ_{ISI}^2 is computed per Equation (93A–31) and Equation (93A–29).
- ~~g)~~ Compute the slope of the pulse response of the victim path $h_j(n)$ as shown in Equation (93A–28). The variance of the amplitude error due to timing jitter σ_j^2 is computed per Equation (93A–32) and Equation (93A–29).
- ~~h)~~ The variance of the amplitude for path k is given by Equation (93A–33) where the phase index m can assume any integer value from 0 to $M-1$. Denote the value of m that maximizes the variance for path k as i . The variance of the amplitude for the combination of all crosstalk paths σ_{XT}^2 is then computed using Equation (93A–34), which is the sum of the maximum variances for the individual paths $k=1$ to $K-1$.

- i) ~~h)~~ Compute the variance of the noise at the output of the receive equalizer σ_N^2 based on the one-sided spectral density η_0 referred to the receiver noise filter input per Equation (93A–35).
- j) ~~i)~~ Compute the figure of merit (FOM) per Equation (93A–36).

Replace Equation (93A–26) in 93A.1.6 with the following equation:

$$b(n) = \begin{cases} bb_{\min}(n) & h^{(0)}(t_s + nT_b)/h^{(0)}(t_s) < bb_{\min}(n) \\ bb_{\max}(n) & h^{(0)}(t_s + nT_b)/h^{(0)}(t_s) > bb_{\max}(n) \\ h^{(0)}(t_s + nT_b)/h^{(0)}(t_s) & \text{otherwise} \end{cases} \quad (93A-26)$$

Replace Equation (93A–27) in 93A.1.6 with the following equation:

$$h_{ISI}(n) = \begin{cases} 0 & n = 0 \\ h^{(0)}(t_s + nT_b) - h^{(0)}(t_s)b(n) & 1 \leq n \leq N_f \\ h^{(0)}(t_s + nT_b) & \text{otherwise} \end{cases} \quad (93A-27)$$

Insert 93A.1.6.1 at the end of 93A.1.6 as follows:

93A.1.6.1 Determination of DFE floating tap locations and magnitudes

The following procedure is used to determine the bank location and tap coefficient magnitudes for each bank of DFE floating taps.

First, the locations of the floating banks are determined using the following steps:

- a) Potential locations for the first tap of a floating bank start at the first post-cursor position after the fixed taps, $N_b + 1$, and end at the specified maximum location minus the number of taps in the bank plus one, $N_f - N_{bf} + 1$. Set the initial values of the provisional coefficients, defined as $b'(n)$, to zero for $N_b + 1 \leq n \leq N_f$.
- b) Define σ_{DFE}^2 to be the sum-of-squares of the tap coefficients for a bank, according to Equation (93A–37a). Calculate σ_{DFE}^2 for each potential bank location, where the coefficient magnitudes are determined using Equation (93A–26). When performing the calculation, if any tap of the floating bank that is being placed overlaps with one or more taps of a previously placed bank, then remove that location from consideration.
- c) Place the floating bank at the location that has the maximum σ_{DFE}^2 . The coefficient magnitudes calculated in b) become the provisional coefficients, $b'(n)$, for the bank, subject to the tail constraints in the steps that follow.
- d) Repeat steps a) through c) for each bank.

After placing the floating banks, the coefficient magnitudes for all floating taps are computed as follows:

- e) For all floating taps that are located between $N_b + 1$ and the start of the ISI tail, $N_{ts} - 1$, set $b(n) = b'(n)$.
- f) For all floating taps that are located between N_{ts} and N_f , compute the tail coefficient root-sum-of-squares, σ_t , which is defined in Equation (93A–37b). Determine the final tap coefficient magnitudes, $b(n)$, by applying the tail weight constraint defined in Equation (93A–37c).

$$\sigma_{DFE}^2 = \sum_{k=1}^{N_{bf}} [b'(N_b + k)]^2 \quad (93A-37a)$$

$$\sigma_t^2 = \sum_{k=N_{ts}}^{N_f} [b'(k)]^2 \quad (93A-37b)$$

$$b(n) = \begin{cases} b'(n) & \sigma_t \leq \sigma_{tmax} \\ b'(n) \frac{\sigma_{tmax}}{\sigma_t} & \sigma_t > \sigma_{tmax} \end{cases} \quad (93A-37c)$$

93A.5 Effective Return Loss (ERL)

Change Table 93A-4 as follows (some unchanged rows not shown):.

Table 93A-4—ERL parameters

Parameter	Reference	Symbol	Units
...			
Target detector error ratio	93A.1.7	DER_0	—
Time-gated propagation delay	93A.5.2	T_{fx}	ns
Tukey window flag	93A.5.1	tw	—

93A.5.1 Pulse time-domain reflection signal

Change the second paragraph of 93A.5.1 as follows:

The filtered return loss, $H_{ii}(f)$, is defined by Equation (93A-58). When the parameter tw is equal to 1, $H_{tw}(f)$ is defined by Equation (93A-58a). When the value of tw is 0 or is not provided by the clause that invokes this method, $H_{tw}(f) = 1$.

Replace Equation (93A-58) with the following equation:

$$H_{ii}(f) = H_i(f) s_{ii}(f) H_r(f) H_{tw}(f) \quad (93A-58)$$

Insert new Equation (93A-58a) and Equation (93A-58b) after Equation (93A-58):

$$H_{tw}(f) = \begin{cases} 1 & f \leq f_r \\ 0.5 \left(1 - \cos \left(\frac{2\pi(f-f_b)}{f_{per}} \right) \right) & f_r < f \leq f_b \\ 0 & f_b < f \end{cases} \quad (93A-58a)$$

$$f_{per} = 2(f_b - f_r) \quad (93A-58b)$$

Change the variable list after Equation (93A-58), now after Equation (93A-58b), in 93A.5.1 as follows:

where

f	is the frequency in GHz
$H_r(f)$	is defined by Equation (93A-20)
$H_t(f)$	is defined by Equation (93A-46)
f_{per}	<u>is defined by Equation (93A-58b)</u>
i	is the port index of the scattering parameters, 1 or 2
f_r	<u>is the receiver 3 dB bandwidth in GHz</u>
$H_{tw}(f)$	<u>is the Tukey window function</u>

93A.5.2 Effective reflection waveform

Change the definition of T_{fx} in the variable list after Equation (93A-62) as follows:

T_{fx}	is twice the propagation delay in ns associated with the test fixture, obtained by measurement or inspection, <u>unless its value is specified by the clause that invokes this method</u>
----------	---